CHAPTER 11

Orthogonality and distances

S48, S51, S54, S56, S60

Before you start · Scalar product and triangles

Without a calculator, in six minutes. Respond briefly. Mobilize Pythagoras, angles and coordinates. Then choose the targeted review indicated under the questions.

1. Calculate $(1;2)\cdot(3;-1)$.
2. Which scalar product characterizes two orthogonal vectors?
3. State Pythagoras.
4. Which unit is used for an area?
5. In an orthonormal benchmark of the plan, calculate the norm of $(3;4)$.
6. Give the cosine of a right angle.

Responses

1 ; 0 ; $c^2 = a^2 + b^2$ in a rectangular triangle; one unit squared; 5; 0.

Targeted review after the diagnostic check

Repeat only the missed questions. For each question: **1.** Multiply Coordinates corresponding and then sum; R60. **2.** Take back the zero product criterion of the plan. **3.** First identify the hypotenuse in a right triangle. **4.** Count square units in a rectangle. **5.** Apply Pythagoras to both coordinates. **6.** Read the right angle point on the circle; R85. R-questions serve as support for the teacher, even if their collective ritual is planned later.

Objectives

Calculate a scalar product; justify an orthogonality; determine a project and a distance; calculate a volume.

Course notes

The co-ordinate calculations of this chapter are done in an orthonormal space coordinate system.

Scalar product in space

For non-zero vectors $\vec{u}, \vec{v}$, $\vec{u}\cdot\vec{v} = \|\vec{u}\|\|\vec{v}\|\cos\theta$ where: $\theta \in [0,\pi]$ is their angle. If one is zero, the scalar product is 0.

$$\vec{u}\cdot\vec{v} = \tfrac{1}{2}(\|\vec{u}+\vec{v}\|^2 - \|\vec{u}\|^2 - \|\vec{v}\|^2).$$

In an orthonormal coordinate system, $\vec{u} = (x, y, z)$ and $\vec{v} = (x', y', z')$,

$$\vec{u} \cdot \vec{v} = xx' + yy' + zz'.$$

The scalar product is symmetrical and bilinear; $\vec{u} \cdot \vec{u} = \|\vec{u}\|^2$. Two vectors are orthogonal if and only if their scalar product is zero.

Example. $\vec{u} = (1, 2, 2)$ and $\vec{v} = (2, -1, 0)$ are orthogonal: $2 - 2 + 0 = 0$. Their norms are 3 and $\sqrt{5}$.

Straight, plan and normal vector

Two lines are orthogonal if their guiding vectors are. A right is orthogonal to a plane if it is orthogonal to two non-collinear directions of this plane. A normal vector to one plane is a non-zero orthogonal vector in all its directions.

Why two directions are enough. If $\vec{n} \cdot \vec{u} = \vec{n} \cdot \vec{v} = 0$, then, for everything $\vec{w} = s\vec{u} + t\vec{v}$, $\vec{n} \cdot \vec{w} = s\vec{n} \cdot \vec{u} + t\vec{n} \cdot \vec{v} = 0$.

Vector condition of belonging

If A belongs to the plan $\mathcal{P}$ of normal vector $\vec{n}$, then $M \in \mathcal{P}$ if and only if $\overrightarrow{AM} \cdot \vec{n} = 0$.

Orthogonal projection and distance

The orthogonal project H of M on a plan is the only point of the plan such as $\overrightarrow{MH}$ or normal in the plan ; if M belongs to the plan, $H = M$. The distance from M on plan east MH. On a director's right $\vec{u}$, the projected verifies $H \in d$ and $\overrightarrow{MH} \cdot \vec{u} = 0$.

Demonstration of minimum. For any point P of the plan, $\overrightarrow{MH} \perp \overrightarrow{HP}$, therefore $MP^2 = MH^2 + HP^2 \geq MH^2$. Equality takes place only for $P = H$. The project is the point of the nearest plan.

Method on a right. If $d = A + \mathbb{R}\vec{u}$, write $H = A + t\vec{u}$ and resolve $(A - M + t\vec{u}) \cdot \vec{u} = 0$. Thus $t = \frac{\overrightarrow{AM} \cdot \vec{u}}{\|\vec{u}\|^2}$.

Angles, areas and volumes

For an angle $\widehat{ABC}$use $\overrightarrow{BA}$ and $\overrightarrow{BC}$: $\cos \widehat{ABC} = \frac{\overrightarrow{BA} \cdot \overrightarrow{BC}}{BA \times BC}$. For the acute angle between two straight lines, take the absolute value of the quotient. The area of a triangle is $\frac{bh}{2}$; the volume of a tetrahedron is $\frac{\mathcal{A}_{\text{base}}\, h}{3}$, where h is an orthogonal distance. Units are squared and cubed respectively.

Geometrical developments and locations

By bilinearity,

$$\|\vec{u} + \vec{v}\|^2 = \|\vec{u}\|^2 + 2\vec{u} \cdot \vec{v} + \|\vec{v}\|^2, \qquad \|\vec{u} - \vec{v}\|^2 = \|\vec{u}\|^2 - 2\vec{u} \cdot \vec{v} + \|\vec{v}\|^2.$$

It is therefore also possible to calculate $\vec{u} \cdot \vec{v} = \frac{1}{2}(\|\vec{u}\|^2 + \|\vec{v}\|^2 - \|\vec{u} - \vec{v}\|^2)$. For two separate points A, B of midpoint I, the locus of the points M such that $MA = MB$ is the plan passing through I normal $\overrightarrow{AB}$: the *plan* Ombudsman. Indeed, by writing vectors since I, developments give $MA^2 - MB^2 = 2\overrightarrow{IM} \cdot \overrightarrow{AB}$. Equality of distances is therefore equivalent to $\overrightarrow{IM} \cdot \overrightarrow{AB} = 0$. **Supplement, without dedicated evaluation in 2026·27.** Two planes are perpendicular when their normal vectors are orthogonal. Do not confuse this property with the parallelism of the plans, which corresponds to normals Collinear.

See a projection, then calculate it

11.1 A scalar product may be negative

First use: S48.

Definition · Scalar product

For two non-zero vectors $\vec{u}, \vec{v}$, $\vec{u} \cdot \vec{v} = \|\vec{u}\| \|\vec{v}\| \cos\theta$ where: $\theta \in [0; \pi]$ is their angle. orthonormal coordinate system, this product is $xx' + yy' + zz'$. Invalidity characterizes the orthogonality of directions.

Solved Example · Recognize an obtuse angle

For $\vec{u} = (1;0;0)$ and $\vec{v} = (-1;1;0)$ the product is -1, norms are 1 and $\sqrt{2}$, therefore $\cos\theta = -1/\sqrt{2}$ and $\theta = 3\pi/4$. A negative product indicates an obtuse or flat angle, not a negative length.

A1 · Immediate application Calculate the angle between $(1;1;0)$ and $(1;0;1)$ in an orthonormal coordinate system. Could we use the same calculation in any database?

11.2. Two directions are required

First use: S51.

Property · Orthogonality Criterion to a Plan

A non-zero vector is normal at a plane if it is orthogonal with two non-directive vectors It is then orthogonal to any combination of these two. vectors, therefore in any direction of the plan.

Example resolved · Test the three conditions

For $\vec{u} = (1;1;0)$, $\vec{v} = (0;1;1)$ and $\vec{n} = (1;-1;1)$, $\vec{n} \cdot \vec{u} = 0$ and $\vec{n} \cdot \vec{v} = 0$. The two directions do not are non-colonial and $\vec{n} \neq 0$, therefore $\vec{n}$ is normal to the plan they lead.

A2 · Counterexample "Being orthogonal to one right of the plane is enough to be normal on the plane." Refute with two perpendicular directions of the horizontal plane. Also explain the difference between orthogonal and perpendicular straight lines.

Example resolved · A locus of equidistant points

Let A=(-1;0;0) and B=(1;0;0). Their midpoint is O. For M=(x;y;z), $MA^2 - MB^2 = (x+1)^2 - (x-1)^2 = 4x$. Thus $MA = MB$ is equal to x=0: the locus is the plan passing through O, normal to $\overrightarrow{AB}$. It is not limited to the mediator drawn in one face.

A3 · Application and justification For A=(0;0;0), B=(2;2;0), describe geometrically the location of the Equidistant points of A and B. Use distance squares to give a relationship between coordinates; find two points of this locus.

11.3 The foot is the nearest point

First use: S54 and S55.

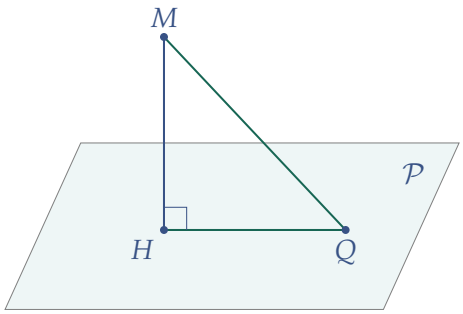


Figure 11.1. Coding represents $MH \perp \mathcal{P}$. For each Q of the plan, $MQ^2 = MH^2 + HQ^2$.

Demonstration. If H is the orthogonal project of M on a plan $\mathcal{P}$, for everything $Q \in \mathcal{P}$, the triangle MHQ is right-angled at H. Thus $MQ^2 = MH^2 + HQ^2 \geq MH^2$. Equality requires $HQ = 0$, therefore $Q = H$. The minimum is reached in a single point. The reasoning is the same for a straight line. □

Solved Example · Projecting on a straight line without plane equation

Let $A = (1;0;0)$, $M = (0;2;3)$ and a right of point A and direction $\vec{u} = (1;1;0)$. Write $\overrightarrow{AH} = t\vec{u}$ gives $H = (1+t;t;0)$. The condition $\overrightarrow{MH} \cdot \vec{u} = 0$ becomes $(1+t) + (t-2) = 0$, therefore $t = 1/2$. Thus $H = (3/2;1/2;0)$ and $MH^2 = 27/2$. Check both membership to the right and orthogonality.

A4 · Partial guidance Project $M = (2;3;1)$ on the right passing through O and direction $(1;1;0)$. Determine project and distance.

A5 · Choice of strategy Project $M = (1;2;3)$ on the plan generated by the $\vec{i}, \vec{j}$. To justify geometrically the foot coordinates, then demonstrate that any other point of the plane is further away.

11.4. The same size by two calculations

First use: S56.

Example Solved · Find a Height

In an orthonormal coordinate system, $O = (0;0;0)$, $A = (6;0;0)$, $B = (0;4;0)$, $C = (0;0;3)$. The triangle OAB is rectangle of area 12 and OC is perpendicular to its plane; the volume is 12 $\times 3/3 = 12$. If the ABC area is 3 $\overline{29}$, its height from O is 3 $V/\mathcal{A}(ABC) = 12/\ \overline{29}$.

A6 · Summary In this tetrahedron, calculate AB, AC and $\overrightarrow{AB} \cdot \overrightarrow{AC}$. Deduce the cosine and then the sine of angle $\widehat{BAC}$, check ABC's area and find the height from O.

A first step: Use $\mathcal{A}(ABC) = \frac{1}{2}$AB AC sin $\widehat{BAC}$. The angle of a triangle has a positive sine.

A7 · Repair an audit A candidate gives a point H of the plan and affirms "it is the project since H What second audit is missing? Give two points in the plan.
$z = 0$ to show that belonging alone is not enough.

Wait a minute at the courthouse. Describe the projection figure by naming the outer point, the plane, the foot, and right angle; explain why distance is not simply calculated at any point of the plan.

Review postponed · S70. Recalculate foot with parametric representation and equation Cartesian; the two geometric conditions remain identical.

Research activity

Nearest straight point · 35 min.

In an orthonormal coordinate system of space, we consider $A(1;0;0)$, $M(0;2;3)$ and the right d passing through A, of a policy vector $\vec{u} = (1;1;0)$. We're looking for the point of d closest to M.

1. Express, based on a real parameter t, the coordinates of a point H of d.

2. Determine the value of t for which $\overrightarrow{MH} \cdot \vec{u} = 0$. Deduce the coordinates of the point H and so obtained.

3. Calculate distance MH.

4. Let P any point of d. Demonstrate that $MP^2 = MH^2 + HP^2$, and then deduce that H is the only point of d closest to M.

Progressive exercises

E1 · Direct application. In an orthonormous base of space, we consider $\vec{u} = (2;-1;3)$ and $\vec{v} = (1;4;0)$. Calculate $\vec{u} \cdot \vec{v}$.

E2 · Application and justification. In an orthonormous basis, we consider $\vec{u} = (a;2;1)$ and $\vec{v} = (3;-1;4)$where: a is real. a so that $\vec{u}$ and $\vec{v}$ be orthogonal.

E3 · Method to be organised. In an orthonormal coordinate system, a plan is directed by $\vec{u} = (1;-1;0)$ and $\vec{v} = (1;0;-1)$. Show that $\vec{n} = (1;1;1)$ He's normal. $A(0;0;0)$ and $B(2;2;2)$. Describe geometrically all the equidistant points of A and B.

E4 · Choice of strategy. In an orthonormous basis, we consider $\vec{u} = (1;0;1)$ and $\vec{v} = (0;1;1)$. Determine the measurement, between 0 and π, of the angle between these two vectors.

E5 · Synthesis or demonstration. A tetrahedron has as its base a right triangle with both sides The height of the tetrahedron relative to this base is 5 cm. Calculate volume in cm ^{3}.

Work in the manual or annal

Barbazo, p. 362, Exercises 23 and 27. In S48: exercise 23, to calculate a scalar product in At home after S54: exercise 27; check the orthogonality of the two control vectors a plan, not just a single zero product.

Duty at home

DM11 · One face and its height · 50 min.

In an orthonormal coordinate system of origin O, the points are considered $A(2;0;0)$, $B(0;2;0)$ and $C(0;0;2)$.

1. Demonstrate that the triangle ABC is equilateral and calculate its area.

2. Show that the vector $\vec{n} = (1;1;1)$ is normal on the plane (ABC).

3. Note H the orthogonal project of O on (ABC). Justify that its coordinates are of the form $(t;t;t)$. Determine t using $\overrightarrow{AH}\cdot\vec{n}=0$.

4. Calculate OH, then the volume of tetrahedron $OABC$ by taking ABC for base. Check this volume by taking OAB for the base.

QCM A · scalar product and orthogonality

Orthonormal coordinate system. After S51; QCM B after S56.

For each question, only one proposal is accurate.
consult the correction.

1. $(1;2;-1)\cdot(2;0;3)$ is valid for:

A. 5 B. -1

C. 1 D. -3

2. The norm of $(2;-1;2)$ is:

A. 9 B. 5

C. $\sqrt{5}$ D. 3

3. To prove that a vector is normal to a plane, simply test its orthogonality at:

A. Two non-collinear directions of the plan B. a direction of the plan

C. Two colinear directions of the plan D. a normal plan

4. Two non-zero vectors of negative scalar product form an angle:

A. Acute B. Law

C. obtuse D. None

5. Two orthogonal lines of space:

A. are still stinging B. may be non-coplanar

C. are parallel D. are confused

MCQ B · projection, minimum and volume

For each question, only one proposal is accurate.
consult the correction.

6. The project $M(2;3;4)$ on the Plan $z=0$ is:

A. $(0;0;4)$ B. $(2;3;4)$

C. $(0;0;0)$ D. $(2;3;0)$

7. H is the project of M on a plan. P of the plan:

A. $MP^2=MH^2+HP^2$ B. $MP=MH+HP$

C. $MH^2=MP^2+HP^2$ D. $MP=HP$

8. A base tetrahedron of floor 6 and height 4 has for volume:

A. 24 B. 12

C. 8 D. 6

9. For vectors u, v units such as $u \cdot v = -\frac{1}{2}$, the angle is:

A. $\pi/3$ B. $2\pi/3$

C. $\pi/2$ D. π

10. $A(1;0;0)$, $M(0;2;3)$, and d of Director $(1;1;0)$ passing through A. The planned M on d is:

A. $(1;1;0)$ B. $(0;2;0)$

C. $(1;0;0)$ D. $(\frac{3}{2};\frac{1}{2};0)$

True or false · expected justification. For each statement, indicate whether it is true or false, and then write evidence or a counter-example.

1. Any point in a plan is closer to M that his project.

2. Two vectors of zero scalar product are colinear.

3. Calculate a volume from two different sides shall give the same result.

Corrected MCQs

1. **B.** $1 \times 2 + 2 \times 0 - 1 \times 3 = -1$. The scalar product is a real, not a vector.

2. **D.** The sum of the squares is 9; the norm is its square root, that is 3.

3. **A.** Two non-colonial directions generate the direction of the plan. directions not perpendicular to the vector.

4. **C.** The angle cosine is negative since the norms are positive. between $\pi/2$ and π.

5. **B.** Orthogonality is about their directions. In addition, perpendicular lines are stinging.

6. **D.** The point belongs to the plane and the vector that connects it to M is vertical, so normal to the plane.

7. **A.** The triangle MHP is right-angled at H where $P \neq H$; · the identity remains valid for $P = H$. It proves $MP \geq MH$.

8. **C.** The volume is one third of the product of the base area by height: 6 $\times 4/3 = 8$.

9. **B.** The scalar product is the angle cosine since the norms are 1; the angle is not oriented $2\pi/3$.

10. **D.** Write $H = A + t(1;1;0)$. Orthogonality $(M - H) \cdot (1;1;0) = 1 - 2t = 0$ given $t = 1/2$.

True/false. False: the projectee achieves the minimum, by Pythagoras. False: $(1;0;0)$ and $(0;1;0)$. True: each height must correspond to the chosen base.

Corrected applications and support

Application 1. Output 1, norms $\sqrt{2}$ and$\sqrt{2}$, therefore cosine $1/2$, angle $\pi/3$. The coordinated formula requires an orthonormal coordinate system.

Application 2. In the plan $z = 0, \vec{\imath}\cdot\vec{\jmath} = 0$, but $\vec{\jmath}$ belongs to the plan and is not normal. orthogonal directions are orthogonal; they are perpendicular if they also cut.

Implementation 4. $H = (t;t;0)$ and $\overrightarrow{MH}\cdot(1;1;0) = 2t-5 = 0$, therefore $H = (5/2;5/2;0)$. $MH^2 = 1/4 + 1/4 + 1 = 3/2$; distance $\sqrt{6}/2$.

Application 5. $H = (1;2;0)$; $\overrightarrow{HM} = (0;0;3)$ is orthogonal in both directions $\vec{\imath},\vec{\jmath}$. For $Q = (x;y;0)$, $MQ^2 = (x-1)^2 + (y-2)^2 + 9 \geq 9$, equality only in H.

Implementation 6. $AB = \sqrt{52}$, $AC = \sqrt{45}$, product 36. Cosine is worth 36 $/\sqrt{2340}$; the sine is worth $\overline{2340 - 1296}/\sqrt{2340} = 6\sqrt{29}/\sqrt{2340}$. Thus, it is 3 $\overline{29}$ and height 12/ $\sqrt{29}$.

Implementation 7. He's missing MH orthogonal plan (or $\overrightarrow{MH}$ colinear to a normal). $M = (1;2;3)$, H=(1;2;0) is suitable but Q=(0;0;0), yet in the plan, is not the projected.

Driving

Search the height in the figure before calculating a volume. For angle calculation, require verification cos $\theta \in [-1;1]$. Keep an exact expression before approximation.

Implementation 3. The midpoint is I=(1;1;0) and a normal vector is (2;2;0). The locus is the plane through I with this Normal. $MA^2 - MB^2 = 4x + 4y - 8$; therefore x+y=2. For example (1;1;0) and (2;0;3) agree. all points of the plan, without restriction on z.

Conduct and correction

Benchmarks for the teacher

Start with the three writings of the scalar product. The orthogonal lines are not necessarily pruning: " perpendicular" requires the intersection. To demonstrate that a right is orthogonal to a plane, choose two non-collinear directions. The general formula of distance point-plane is not required by heart: prefer the projected. Cartesian equations used in S54 progression can be announced by vector condition, then Formalised in S67.

Corrected activity

1. One item H of d written $H = A + t\vec{u}$, therefore $H(1+t, t, 0)$ for $t \in \mathbb{R}$.

2. $\overrightarrow{MH} = (1+t, t-2, -3)$. The scalar product with $\vec{u} = (1,1,0)$ value $1 + t + t - 2 = 2t - 1$. Sound cancellation given $t = \frac{1}{2}$ and $H(\frac{3}{2}, \frac{1}{2}, 0)$.

3. The coordinate system shall be orthonormal: $MH^2 = (\frac{3}{2})^2 + (-\frac{3}{2})^2 + (-3)^2 = \frac{27}{2}$. So $MH = \frac{3\sqrt{6}}{2}$.

4. For $P \in d$, $\overrightarrow{HP}$ is colinear to $\vec{u}$ and thus orthogonal to $\overrightarrow{MH}$. The theorem of Pythagoras gives $MP^2 = MH^2 + HP^2 \geq MH^2$, with equality only if $P = H$. The calculated point is very unique point of the nearest right to M.

E1. In the orthonormal coordinate system, the scalar product is $2 \times 1 + (-1) \times 4 + 3 \times 0 = -2$.

E2. The scalar product is $3\ a + 2(-1) + 1 \times 4 = 3a + 2$. Orthogonality is equivalent to its cancellation: $3a + 2 = 0$, therefore $a = -\frac{2}{3}$.

E3. The two directors are non-collinear and the two scalar products are zero. $I(1,1,1)$. The location sought is the plan passing through I and normal to $\overrightarrow{AB} = (2,2,2)$, so normal $(1,1,1)$.

E4. The two vectors are non-zero, norms $\sqrt{2}$, and their scalar product is 1. So their angle $\theta \in [0, \pi]$ check $\cos\ \theta = \frac{1}{\sqrt{2}\sqrt{2}} = \frac{1}{2}$either $\theta = \frac{\pi}{3}$.

E5. The area of the base rectangle is $\frac{3\times 4}{2} = 6\ \text{cm}^2$. The height perpendicular to this base is 5 cm, so $V = \frac{6\times 5}{3} = 10\ \text{cm}^3$.

Corrected from DM11

1. $\overrightarrow{AB} = (-2,2,0)$, $\overrightarrow{AC} = (-2,0,2)$ and $\overrightarrow{BC} = (0,-2,2)$; The three norms are worth $2\sqrt{2}$. The triangle is Its height is equal to $\sqrt{(2\sqrt{2})^2 - (\sqrt{2})^2} = \sqrt{6}$; its area is therefore $\frac{(2\sqrt{2})(\sqrt{6})}{2} = 2\sqrt{3}$.

2. The vectors $\overrightarrow{AB}, \overrightarrow{AC}$ their scalar products with $(1,1,1)$ value respectively $-2 + 2 = 0$ and $-2 + 2 = 0$. The non-zero vector $\vec{n}$ is therefore normal to (ABC).

3. The project O belongs to the right of director $\vec{n}$ passing through O, therefore $H = (t,t,t)$. Its condition of membership in the Eastern Plan $(t-2) + t + t = 0$, either $t = \frac{2}{3}$. We get $H(\frac{2}{3}, \frac{2}{3}, \frac{2}{3})$, which verifies the two conditions of a project.

4. $OH = \sqrt{3(\frac{2}{3})^2} = \frac{2}{\sqrt{3}}$. Taking the base ABC, volume is $V = \frac{(2\sqrt{3})(\frac{2}{\sqrt{3}})}{3} = \frac{4}{3}$ units of volume. Other base: the triangle OAB is right-angled at O, d ·aire $\frac{2\times 2}{2} = 2$; the right (OC) is perpendicular on the plan (OAB), and $OC = 2$. Thus $V = \frac{2\times 2}{3} = \frac{4}{3}$.

Corrected work in the manual or an anal

Exercise 23. In the given orthonormal coordinate system, the scalar product is the sum of the products corresponding coordinates:

$$(-3{,}5)(-4) + 2 \times 5{,}5 + (-4)(-3) = 14 + 11 + 12 = 37,$$

$$\frac{2}{3}\left(-\frac{3}{4}\right) + \frac{15}{7}\frac{14}{3} + \left(-\frac{12}{11}\right)\left(-\frac{22}{4}\right) = -\frac{1}{2} + 10 + 6 = \frac{31}{2}.$$

Exercise 27. In each case, the two vectors $\vec{u}$ and $\vec{v}$ given lead the plan and are not
A non-zero vector $\vec{w}$ is normal if, and only if, it is orthogonal to each of them.

1. $\vec{w} \cdot \vec{u} = -132 + 312 - 180 = 0$ and $\vec{w} \cdot \vec{v} = -66 + 546 - 480 = 0$. The vector $\vec{w}$ is normal in the plan.
2. $\vec{w} \cdot \vec{u} = 20 + 6 - 26 = 0$, but $\vec{w} \cdot \vec{v} = 18 - 6 - 10 = 2 \neq 0$. It is not normal in the plan.
3. $\vec{w} \cdot \vec{u} = \frac{1}{2} + 6 - 7 = -\frac{1}{2} \neq 0$. This single non-zero product is sufficient to exclude normality.